

PROFESSIONAL SUMMARY

Entry-level Data Analyst with hands-on experience in data cleaning, statistical analysis, visualization, and reporting using SQL, Python, R, Tableau, and Power BI. Skilled in transforming raw datasets into actionable insights through exploratory analysis and storytelling. Experienced in working with research and operational datasets during internship and attachment programs. Passionate about solving business and social problems using data-driven solutions.

SKILLS

- Programming & Querying: SQL, Python, R
- Data Visualization: Tableau, Power BI, Excel, Data Storytelling
- Statistical Analysis: Hypothesis Testing, Mixed-Effects Modeling, Exploratory Data Analysis (EDA), Predictive Modeling
- Data Engineering: Google BigQuery, Excel Power Query, GitHub Actions
- Reporting & Documentation: Quarto, Microsoft Word, Research Analytics
- Database Management: PostgreSQL, Data Cleaning

PROJECTS

HOW KENYAN HOUSEHOLDS CAN CUT MAIZE FLOUR COSTS – Personal Project – Kisumu, Kenya, KE Aug 2025

- Cleaned and analyzed 10 years of dry maize vs. maize flour price data using SQL, Python, and R to test if milling saves money.
- Found milling dry maize (**KSh130/2kg**) saves households **KSh40 per 2kg** compared to buying packaged flour (**KSh170/2kg**), if milling losses are ignored.
- Showed that with **25% milling losses**, the effective cost rises to **KSh174/2kg**, making packaged flour cheaper at retail.
- Confirmed differences with Wilcoxon tests; cost gap of **KSh5 per 2kg** was statistically significant though effect size was modest.
- **Outcome:** While milling can cut costs, savings quickly disappear once milling inefficiencies are factored in.

WHAT DRIVES NAIROBI RENTAL PRICES – Personal Project – Kisumu, Kenya, KE Aug 2026

- Built an automated pipeline scraping 96,853+ Property24 Kenya rental listings over 35+ days with zero missed runs, storing data in **BigQuery** and **PostgreSQL**.
- Fit a **linear mixed-effects model** (R, lme4/lmerTest) on 15,302 cleaned listings, using location and agency as crossed random effects.
- Found an ICC of 0.595 — location and agency explain nearly 60% of price variation before individual property features are considered.
- Showed floor size had minimal effect on price (1.05x multiplier per log unit), while furnishing status (1.60x) and bathroom count (1.22x per bathroom) were far stronger drivers.
- Outcome: demonstrated that conventional size-driven pricing heuristics do not hold in the Nairobi rental market.

WORK EXPERIENCE

DATA ANALYST – West Kenya Sugar Company Limited (Olepito Unit) – Busia, Kenya, KE June 2026 – Present

- Supporting the **DigiCane** agriculture management system at the **Olepito Unit**, including data quality checks, cane variety and acreage tracking, and operational reporting.
- Maintaining data integrity and system operations while supporting reporting and process compliance across agriculture workflows.
- Participating in a structured mentorship program under the Agriculture Admin Manager to build expertise in agricultural systems administration.

DATA ANALYTICS TUTOR – TeacherOn – Remote Jan 2026 – June 2026

- Tutor students worldwide in data analysis, statistics, research methods, and analytical problem-solving, covering SQL, Excel, Power BI, **R**, and **Python** from beginner to advanced level.
- Break down regression models, **hypothesis testing**, and research interpretation into clear, step-by-step explanations tailored to each student's background and goals.
- Guide learners through data pipelines, workflow automation, and data visualization projects, building their confidence with analytical tools while meeting academic standards.

- Completed a 4-week virtual internship analyzing student engagement datasets to uncover trends and behavioral patterns using **Python, SQL**, and machine learning workflows.
- Performed data preprocessing, EDA, and built predictive models to identify factors driving sign-ups, completions, and drop-offs; generated actionable insights through clear visualizations and reports.
- Recognized as a **Star Performer** for outstanding analytical work and solution-oriented insights.

- Supported the PRISMA research study through data collection, entry, and validation activities across laboratory and research teams.
- Participated in team-based data entry and dataset management to improve consistency and completeness of research records.
- Cleaned and analyzed health research data using **Python, R, Stata**, and **SQL** to support statistical analysis and reporting.
- Developed reports and data visualizations to communicate research findings and study progress to stakeholders.

EDUCATION

CERTIFICATIONS

Google Data Analytics Professional Certificate

- Completed 8 courses covering: data foundations, data cleaning, analysis, visualization, R programming, and SQL.
- Developed hands-on skills in preparing, processing, analyzing, and sharing data for decision-making.
- Gained proficiency in tools including Excel, SQL, Tableau, and R.
- Verified at: <https://coursera.org/verify/professionalcert/JJHO687M0ZX0>

ACHIEVEMENTS

- **Large-Scale Workflow Automation Deployment** – Successfully deployed and monitored 14,118+ automated workflow runs using GitHub Actions to extract, process, and upload agricultural market data into Google BigQuery, demonstrating strong system architecture, automation, and infrastructure management skills.
- **Anomaly Detection in M-PESA Transactions** – Developed a three-stage anomaly detection framework on 5,126+ personal M-PESA transaction records, reducing flagged anomalies from 496 (IQR) to 169 (STL) and 49 (Isolation Forest), demonstrating expertise in statistical modeling, time series decomposition, and context-aware anomaly detection.
- **Star Performer – Excelerate** – Recognized as a Star Performer during the Excelerate Data Analyst Internship for outstanding analytical work and solution-oriented insights.

REFERENCES

Dr Fredrick Onduru – Statistician – KEMRI-CHGR
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